Duration: You will need at least two hours for this exam. I recommend planning for four hours.

DO NOT OPEN the exam until you are ready to work. Include COMPLETE and DETAILED JUSTIFICATIONS for all the problems.

This exam is due no later than the start of our final exam on Wednesday, May 11th, 2022

You may not consult in any form with any other person while doing this take-home exam. Do not consult any online sources. You may, however, consult our textbook, your class notes, your own homework, and resources I have posted or given out in class. **You are welcome to email me if you need clarifications on any of the problems.**

After you have completed the exam, you should do the following:

1. Recopy your final solutions neatly, *starting each problem on a new page.*

2. Print your name at the top of this page, and read and sign the certification below.

3. Include this cover sheet on top of your rewritten solutions. (Your solutions will not be graded without your signed cover sheet.)

CERTIFICATION:

Please read and sign (certify that you agree to and followed) the following statement before submitting your solutions to this exam.

I hereby certify that this exam represents my work, and my work only, and that I did not consult with any other person in answering the questions on this exam. I have consulted only with our textbook, my class notes, and my own homework. I have not consulted with any other resources.

Signature:

Problem	Score
1 *(10 pts)*	
2 *(10 pts)*	
3 *(15 pts)*	
4 *(10 pts)*	
5 *(10 pts)*	
6 *(10 pts)*	
7 *(10 pts)*	
8 *(10 pts)*	
TOTAL*(85 pts)*	

1. Show whether the following PDEs are linear using the principle of superposition for two solutions, u_1 and u_2: Explain why or why not.
   (a) $u_x + 2u_{tt} = 2$
   (b) $u_x - x^2 u_{tt} = 0$
2. Solve $2u_x + 5u_y = 0$, given condition $u(1, y) = e^y$ using method of characteristics.
3. Set up and solve the homogeneous heat/diffustion initial-boundary-value problem for a rod with a length of $5cm$, specific heat of $0.15cal/(g \cdot^\circ C)$, a density of $3g/(cm^3)$, and thermal conductivity of $0.75cal/(cm \cdot sec \cdot^\circ C)$ with the left end in a bath at $20°C$, and the right end insulated ($h = 0.5cal/(cm^2 \cdot sec \cdot^\circ C)$, and an initial fixed temperature of $65°C$.
4. Solve the 2D wave equation, $u_{tt} = 4\nabla^2 u$ on the rectangle $0 \le x \le 2$, $0 \le y \le 3\pi$, given boundary conditions $u_x(0, y, t) = u_x(2, y, t) = u(x, 0, t) = u(x, 3\pi, t) = 0$, with initial conditions $u(x, y, 0) = 2\cos(\pi x)\sin(\frac{y}{3}) - 9\cos(\frac{\pi x}{2})\sin(y)$, and $u_t(x, y, 0) = 0$
5. Solve the Laplace equation using separation of variables $u_{xx} + u_{yy} = 0$, on $0 \le x \le \pi$, $0 \le y \le 2$, given conditions $u(0, y) = u(\pi, y) = 0$, and $u(x, 0) = 5\sin(8x)$, $u(x, 2) = \sin(x)$
6. Solve the heat equation using Fourier transforms, $u_t = 2u_{xx}$, on an infinite domain with initial condition
   $$u(x, 0) = \begin{cases} 1, & |x| \le 1 \\ 0, & \text{otherwise} \end{cases}$$
7. Convert $u_{xy}(x, y)$ to polar coordinates. Show ALL work.
8. Solve the Dirichlet problem on a disk: $\nabla^2 u = 0$, on $0 \le r \le 4$, given boundary condition $u(4, \theta) = 5$ (assume periodic boundary conditions in θ, it is a disk...)

	$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(\alpha) e^{i\alpha x}\, d\alpha$		$F(\alpha) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\alpha x}\, dx$
1.	$f'(x)$	1.	$i\alpha F(\alpha)$
2.	$xf(x)$	2.	$iF'(\alpha)$
3.	$f(x-c)$	3.	$e^{-ic\alpha} F(\alpha)$
4.	$e^{icx} f(x)$	4.	$F(\alpha - c)$
5.	$f(cx)$	5.	$\frac{1}{c} F\left(\frac{\alpha}{c}\right)$
6.	$\frac{1}{c} f\left(\frac{x}{c}\right)$	6.	$F(c\alpha)$
7.	$f(x) * g(x)$	7.	$\sqrt{2\pi}\, F(\alpha) G(\alpha)$
8.	$f(x) g(x)$	8.	$\frac{1}{\sqrt{2\pi}}\, F(-\alpha) * G(-\alpha)$
9.	$e^{-c\lvert x\rvert}$	9.	$\sqrt{\frac{2}{\pi}}\, \frac{c}{\alpha^2 + c^2}$
10.	$\frac{1}{x^2+c^2}$	10.	$\sqrt{\frac{\pi}{2}}\, e^{-c\lvert\alpha\rvert}/c$
11.	$\exp[-(x-m)^2/2\sigma^2]/\sigma\sqrt{2\pi}$	11.	$\exp\left[-\frac{\sigma^2\alpha^2}{2} - im\alpha\right] \Big/ \sqrt{2\pi}$
12.	$\begin{cases} 1, & \text{if } \lvert x\rvert \le L, \\ 0, & \text{if } \lvert x\rvert > L. \end{cases}$	12.	$\sqrt{\frac{2}{\pi}}\, \frac{\sin L\alpha}{\alpha}$
13.	$\frac{\sin Lx}{x}$	13.	$\begin{cases} \sqrt{\frac{\pi}{2}}, & \text{if } \lvert\alpha\rvert \le L, \\ 0, & \text{if } \lvert\alpha\rvert > L. \end{cases}$
14.	$\delta(x)$	14.	$\frac{1}{\sqrt{2\pi}}$
15.	1	15.	$\sqrt{2\pi}\, \delta(\alpha)$
16.	$H(x)$	16.	$\frac{1}{i\alpha\sqrt{2\pi}} + \sqrt{\frac{\pi}{2}}\, \delta(\alpha)$
17.	$\operatorname{sgn} x$	17.	$\sqrt{\frac{2}{\pi}}\, \frac{1}{i\alpha}$
18.	$\frac{1}{x}$	18.	$\sqrt{\frac{\pi}{2}} \frac{\operatorname{sgn} \alpha}{i}$
19.	$\operatorname{sgn} x * f(x)$	19.	$\frac{2F(\alpha)}{i\alpha}$
20.	$\frac{f(x)}{x}$	20.	$-\frac{i}{2} \operatorname{sgn} \alpha * f(-\alpha)$